

Programming in R

Summary of Syntaxes and Commands

(Summary Statistics and Data Visualization)

Contents

1	Creating Objects	3
2	Creating a Numeric Vector	3
3	Displaying Data	3
4	Measures of Location	4
5	Measures of Spread	5
6	Measures of Shape	5
7	Measures of Peakedness	6
8	Bar Plot	6
9	Scatter Plot	6
10	Line Plot	7
11	Histogram	7
12	Box Plot	7
13	Horizontal Box Plot	7
14	Dot Chart	8
15	Adding a Horizontal Line	8
16	Adding Text to Graph	8
17	Density Curve	9
18	Drawing Density Curve	9
19	Useful Plot Parameters	9

20 Common Colours	10
21 Summary Table of Commands	10

1 Creating Objects

Syntax

```
1 object_name <- value
```

Example

```
1 marks <- c(45,52,60,70,85,90,78,65)
```

2 Creating a Numeric Vector

Syntax

```
1 vector_name <- c(value1, value2, value3, ...)
```

Example

```
1 marks <- c(45,52,60,70,85,90,78,65)
```

3 Displaying Data

print()

Displays the complete object.

```
1 print(object)
```

Example

```
1 print(marks)
```

cat()

Prints text and values together.

```
1 cat("Mean =", mean(marks))
```

4 Measures of Location

Mean

Syntax

```
1 mean(x)
```

Example

```
1 mean(marks)
```

Median

Syntax

```
1 median(x)
```

Example

```
1 median(marks)
```

Summary

Displays

- Minimum
- First Quartile
- Median
- Mean
- Third Quartile
- Maximum

Syntax

```
1 summary(x)
```

Example

```
1 summary(marks)
```

5 Measures of Spread

Range

```
1 range(x)
```

Example

```
1 range(marks)
```

Variance

```
1 var(x)
```

Example

```
1 var(marks)
```

Standard Deviation

```
1 sd(x)
```

Example

```
1 sd(marks)
```

Interquartile Range

```
1 IQR(x)
```

Example

```
1 IQR(marks)
```

6 Measures of Shape

Install package

```
1 install.packages("moments")
```

Load package

```
1 library(moments)
```

Skewness

```
1 skewness(x)
```

Example

```
1 skewness(marks)
```

7 Measures of Peakedness

Kurtosis

```
1 kurtosis(x)
```

Example

```
1 kurtosis(marks)
```

8 Bar Plot

Syntax

```
1 barplot(height ,  
2         main="Title",  
3         xlab="X-axis",  
4         ylab="Y-axis",  
5         col="color")
```

Example

```
1 barplot(marks ,  
2         col="skyblue",  
3         main="Students Marks")
```

9 Scatter Plot

Syntax

```
1 plot(x)
```

Example

```
1 plot(marks)
```

10 Line Plot

Syntax

```
1 plot(x,  
2     type="o",  
3     pch=16,  
4     col="blue")
```

Example

```
1 plot(marks ,  
2     type="o",  
3     pch=16,  
4     col="darkgreen")
```

11 Histogram

Syntax

```
1 hist(x,  
2     main="Title",  
3     xlab="Label",  
4     col="color")
```

Example

```
1 hist(marks ,  
2     col="lightblue")
```

12 Box Plot

Syntax

```
1 boxplot(x,  
2     main="Title",  
3     col="color")
```

Example

```
1 boxplot(marks ,  
2     col="orange")
```

13 Horizontal Box Plot

Syntax

```
1 boxplot(x,  
2       horizontal=TRUE)
```

Example

```
1 boxplot(marks,  
2       horizontal=TRUE)
```

14 Dot Chart

Syntax

```
1 dotchart(x,  
2       pch=19,  
3       col="blue")
```

Example

```
1 dotchart(marks)
```

15 Adding a Horizontal Line

Syntax

```
1 abline(h=value,  
2       col="color",  
3       lwd=2)
```

Example

```
1 abline(h=mean(marks),  
2       col="red")
```

16 Adding Text to Graph

Syntax

```
1 text(x, y, labels)
```

Example

```
1 text(6,  
2     mean(marks)+2,  
3     "Mean")
```


17 Density Curve

Syntax

```
1 density(x)
```

Example

```
1 density(marks)
```

18 Drawing Density Curve

Syntax

```
1 lines(density(x),  
2       col="red",  
3       lwd=2)
```

Example

```
1 hist(marks,  
2       probability=TRUE)  
3  
4 lines(density(marks),  
5       col="red",  
6       lwd=2)
```

19 Useful Plot Parameters

Parameter	Purpose
main	Graph title
xlab	X-axis label
ylab	Y-axis label
col	Colour
pch	Plotting symbol
type	Type of plot
lwd	Line width
horizontal	Horizontal box plot
probability	Density scale histogram

20 Common Colours

Colour Name	Appearance
red	Red
blue	Blue
green	Green
darkgreen	Dark Green
orange	Orange
pink	Pink
skyblue	Sky Blue
lightblue	Light Blue
lightgreen	Light Green
lightyellow	Light Yellow
gray	Gray
black	Black

21 Summary Table of Commands

Command	Purpose
<code>c()</code>	Create vector
<code><-</code>	Assignment operator
<code>print()</code>	Display object
<code>cat()</code>	Print formatted output
<code>mean()</code>	Mean
<code>median()</code>	Median
<code>summary()</code>	Five-number summary
<code>range()</code>	Minimum and maximum
<code>var()</code>	Variance
<code>sd()</code>	Standard deviation
<code>IQR()</code>	Interquartile range
<code>library()</code>	Load package
<code>install.packages()</code>	Install package
<code>skewness()</code>	Skewness
<code>kurtosis()</code>	Kurtosis
<code>barplot()</code>	Bar chart
<code>plot()</code>	Scatter/line plot
<code>hist()</code>	Histogram
<code>boxplot()</code>	Box plot
<code>dotchart()</code>	Dot chart
<code>density()</code>	Density estimation
<code>lines()</code>	Add line to graph
<code>abline()</code>	Horizontal/vertical line
<code>text()</code>	Add text to graph